



Fire Extinguishers: Types, Uses, and Safety

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What is Fire



Fire is a very rapid chemical reaction between oxygen and a combustible material, which results in the release of heat, light, flames, and smoke.



Fire Triangle



For fire to exist, the following four elements must be present at the same time: :

- Enough oxygen to sustain combustion.
- Enough heat to raise the material to its ignition temperature.
- Some sort of fuel or combustible material.
- The chemical reaction that is fire.





Stages of Fire



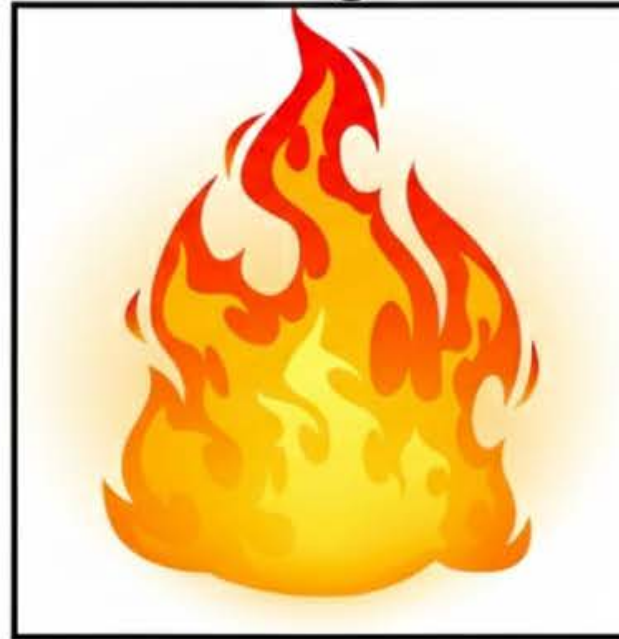
**Incipient
Stage**



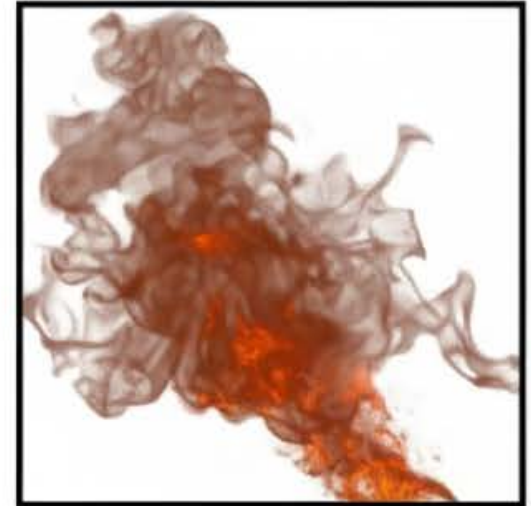
**Growth
Stage**



**Fully Developed
Stage**



**Decay
Stage**



Methods of Extinguishing a Fire



Every method works by removing one side of the fire triangle:



Cooling — Removes Heat

- Lowers the fuel's temperature below its ignition point.
- Example: water, CO₂ dry-ice discharge.



Smothering — Removes Oxygen

- Cuts off the air supply that feeds the flame.
- Example: foam, fire blanket, CO₂ gas.



Starvation — Removes Fuel

- Takes away or isolates the burning material.
- Example: shutting off a gas valve, clearing nearby fuel.



Types of Fire



Class A : Ordinary Combustibles

- **Fuel:** Wood, paper, cloth, rubber, plastics, trash.
- **Characteristics:** Leaves ash after burning.
- **Extinguishing Method:** Water, foam, or dry chemical.



Class B : Flammable Liquids & Gases

- **Fuel:** Petrol, diesel, oil, kerosene, alcohol, propane, LPG,
- **Characteristics:** Spreads quickly, produces intense flames.
- **Extinguishing Method:** Foam, DCP, (never use water).

Types of Fire



Class C : Electrical Fires

- **Fuel:** Electrical equipment (wiring, circuit breakers, motors).
- **Characteristics:** Caused by electrical faults.
- **Extinguishing Method:** CO₂, DCP



Class D : Combustible Metals

- **Fuel:** Magnesium, sodium, potassium, aluminum powder.
- **Characteristics:** Burns at very high temperature, reacts violently with water.
- **Extinguishing Method:** Special dry powder

Types of Fire



Class K : Cooking Oils & Fats

- **Fuel:** Vegetable oils, animal fats, deep fryers.
- **Characteristics:** Very high ignition temperature, re-ignition risk.
- **Extinguishing Method:** Wet chemical extinguishers (creates a soapy foam layer).

Fire Classification & Extinguisher Table



		Class A	Class B	Class C	Class D	Electrical	Class F
Type	Fire	Combustible Materials (e.g. paper & wood)	Flammable Liquids (e.g. paint & petrol)	Flammable Gases (e.g. butane & methane)	Flammable Metals (e.g. lithium & potassium)	Electrical Equipment (e.g. computers & generators)	Deep Fat Fryers (e.g. chip pans)
Extinguisher	Water	✓	✗	✗	✗	✗	✗
	Foam	✓	✓	✗	✗	✗	✗
	Dry Powder	✓	✓	✓	✓	✓	✗
	CO ₂	✗	✓	✗	✗	✓	✗
	Wet Chemical	✓	✗	✗	✗	✗	✓

CO2 Fire Extinguisher



Content: Filled with CO₂, a non-flammable gas under extreme pressure

How it works:

- Displaces oxygen, removing the oxygen element of the fire triangle.
- High-pressure discharge creates dry ice, which also cools the fire

Suitable for: Class B (flammable liquids) and Class C (electrical) fires only

Important:

- CO₂ is not recommended for Class A fires because they may continue to smolder and re-ignite after the CO₂ dissipates.
- Never use CO₂ extinguishers in a confined space while people are present without proper respiratory protection.



Smoldering Fire



Multi-purpose - Dry Chemical Extinguishers



Content: Dry Chemical fire extinguisher

How it works:

- Coats fuel with a thin layer of fire-retardant powder
- Separates fuel from oxygen

Suitable for: Usually rated for Class B and C fires. Some are multi-purpose (ABC) for Class A, B, and C fires

Important:

- May leave a powder residue that can damage sensitive equipment or electronics.
- After use on Class A fires, monitor the area as smoldering may continue.



Fire Extinguisher Use



Most fire extinguishers operate using the following P.A.S.S. technique:

- 1. PULL:** Pull the pin. This will also break the tamper seal.
- 2. AIM:** Aim low, pointing the extinguisher nozzle (or its horn or hose) at the base of the fire.

NOTE: Do not touch the plastic discharge horn on CO2 extinguishers, it gets very cold and may damage skin.
- 3. SQUEEZE:** Squeeze the handle to release the extinguishing agent.
- 4. SWEEP:** Sweep from side to side at the base of the fire until it appears to be out. Watch the area. If the fire re-ignites, repeat steps 2 - 4



Fire Extinguisher Use

HOW TO USE A FIRE EXTINGUISHER



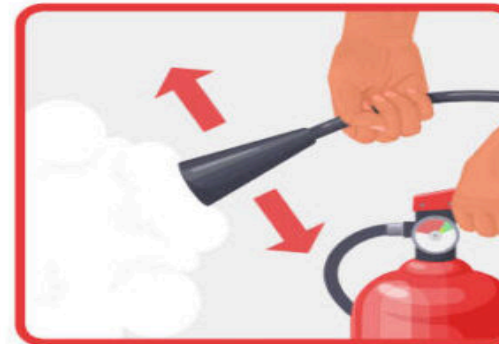
PULL SAFETY PIN



AIM AT THE BASE OF FIRE



SQUEEZE THE LEVER



SWEEP SIDE TO SIDE

Thank you for your attention
&
support

